

## **Car Customization Webpage – A WebBased Vehicle Customization Platform**

### **1. Introduction**

With the growth of digital platforms in the automobile industry, customers increasingly expect interactive tools to visualize and customize vehicles before purchase. Traditional methods of car selection lack flexibility and real-time customization. This project aims to develop a web-based car customization system that allows users to select car models, colors, wheels, interiors, and accessories through an interactive interface. The system provides instant updates of selected features and total price, improving user engagement and decision-making.

### **2. Objectives**

- To develop an interactive web-based car customization application
- To allow users to select and customize different car features
- To dynamically calculate and display the total price
- To implement client–server communication using Node.js and Express.js

### **3. Scope of the Project**

- Selection of car models and variants
- Customization of colors, wheels, interiors, and accessories
- Real-time price calculation based on selected options
- Server-side handling of customization data
- Future potential: database integration, user accounts, and booking system

### **4. Problem Statement**

Customers often find it difficult to visualize customized car options using traditional showroom methods. Manual calculations and limited display options reduce user experience and efficiency. This project provides a centralized web-based platform that allows users to customize cars digitally, view instant updates, and understand pricing clearly, thereby enhancing usability and reducing manual effort.

## 5. Methodology / Technologies

- **Frontend:** HTML, CSS, JavaScript
- **Backend:** Node.js with Express.js
- **Database:** MongoDB

## 6. How It Works (User Perspective)

When a user visits the car customization webpage, they are presented with various car models and customization options. The user selects preferred features such as color, wheels, interior type, and accessories. JavaScript dynamically updates the displayed car details and total price. Once customization is complete, the selected data is sent to the server using Express.js for processing and confirmation. The user can view a final summary of the customized car.

## 7. Expected Outcomes

- Fully functional car customization web application
- Improved understanding of frontend and backend integration
- Real-time customization and pricing system
- Enhanced user interaction and visualization
- Practical exposure to full-stack web development

## 8. Limitations

- Limited to predefined car models and customization options
- No database integration in the initial version
- No payment gateway or booking system included
- Basic visual representation instead of 3D modeling